

# Chapter 1: Introduction

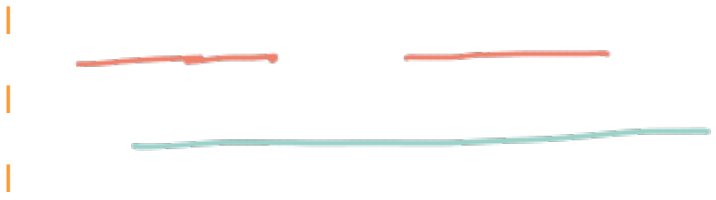
Database System Concepts, 6th Ed

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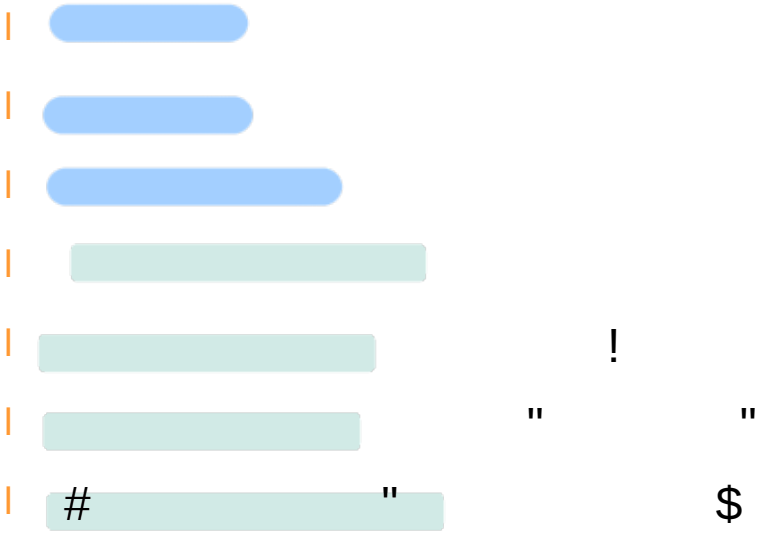


# Database Management System (DBMS)

n



n




n

"

n



# University Database Example

n $

- %
- &
- '() *

n + " " % " ,

"



# Drawbacks of using file systems to store data

| " "

, -

| . "

/ % % " %

| 0 ,

| + "

+ " ' 1 2*

3 4 $ "

# % $



# Drawbacks of using file systems to store data (Cont.)

- "
  - 5 " %
  - 6$ 7
- "
  - 6$ 7% ' " 822* "
    % % " ' " 92 *
- "
  - #

**Database systems offer solutions to all the above problems**



# Data Models





# Relational Model

- n & ' >*
- n 6$

| *ID* | *name* | *dept_name* | *salary* |
|---|---|---|---|
| 22222 | Einstein | Physics | 95000 |
| 12121 | Wu | Finance | 90000 |
| 32343 | El Said | History | 60000 |
| 45565 | Katz | Comp. Sci. | 75000 |
| 98345 | Kim | Elec. Eng. | 80000 |
| 76766 | Crick | Biology | 72000 |
| 10101 | Srinivasan | Comp. Sci. | 65000 |
| 58583 | Califieri | History | 62000 |
| 83821 | Brandt | Comp. Sci. | 92000 |
| 15151 | Mozart | Music | 40000 |
| 33456 | Gold | Physics | 87000 |
| 76543 | Singh | Finance | 80000 |

& %

(a) The *instructor* table



# A Sample Relational Database

| *ID* | *name* | *dept_name* | *salary* |
|---|---|---|---|
| 22222 | Einstein | Physics | 95000 |
| 12121 | Wu | Finance | 90000 |
| 32343 | El Said | History | 60000 |
| 45565 | Katz | Comp. Sci. | 75000 |
| 98345 | Kim | Elec. Eng. | 80000 |
| 76766 | Crick | Biology | 72000 |
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| 58583 | Califieri | History | 62000 |
| 83821 | Brandt | Comp. Sci. | 92000 |
| 15151 | Mozart | Music | 40000 |
| 33456 | Gold | Physics | 87000 |
| 76543 | Singh | Finance | 80000 |

(a) The *instructor* table

| *dept_name* | *building* | *budget* |
|---|---|---|
| Comp. Sci. | Taylor | 100000 |
| Biology | Watson | 90000 |
| Elec. Eng. | Taylor | 85000 |
| Music | Packard | 80000 |
| Finance | Painter | 120000 |
| History | Painter | 50000 |
| Physics | Watson | 70000 |

(b) The *department* table



# Data Definition Language (DDL)

n , ,

6$ **create table** '

**char**'9*

**varchar**'>2*,

**varchar**'>2*

**numeric**'? >**

n =

n " ' *

|

| + "

) " " '+ @ " , *

& " ' **references** A=*

"

| !



# SQL

- **SQL** % " :
  - 6$ 5 % + >>>>>
    **select**
    **from**
    **where** B C>>>>>D
  - 6$ 5 + ) "
    **select**
    **from**
    **where** **and**
    B C) " D

- "
  - = $ % A=
  - ' EF * % % A=
    @
- G H 9



# Database Design?

n + " % l

| *ID* | *name* | *salary* | *dept_name* | *building* | *budget* |
|---|---|---|---|---|---|
| 22222 | Einstein | 95000 | Physics | Watson | 70000 |
| 12121 | Wu | 90000 | Finance | Painter | 120000 |
| 32343 | El Said | 60000 | History | Painter | 50000 |
| 45565 | Katz | 75000 | Comp. Sci. | Taylor | 100000 |
| 98345 | Kim | 80000 | Elec. Eng. | Taylor | 85000 |
| 76766 | Crick | 72000 | Biology | Watson | 90000 |
| 10101 | Srinivasan | 65000 | Comp. Sci. | Taylor | 100000 |
| 58583 | Califieri | 62000 | History | Painter | 50000 |
| 83821 | Brandt | 92000 | Comp. Sci | Taylor | 100000 |
| 15151 | Mozart | 40000 | Music | Packard | 80000 |
| 33456 | Gold | 87000 | Physics | Watson | 70000 |
| 76543 | Singh | 80000 | Finance | Painter | 120000 |


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# Design Approaches



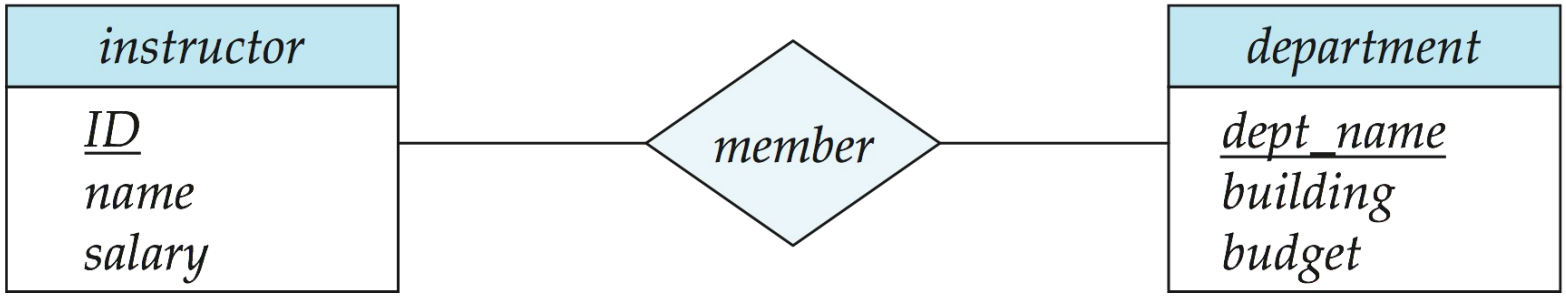
# The Entity-Relationship Model

n

l 6 " 3 4 3 ; 4
;
"

l &

n & " "




**What happened to dept_name of instructor and student?**



# Storage Management

- **Storage manager**
  - %  %:
  - @  "
- 7  %
  - \+  %  ,
  - 6.
- \+
  - 5  !
  - \+ $

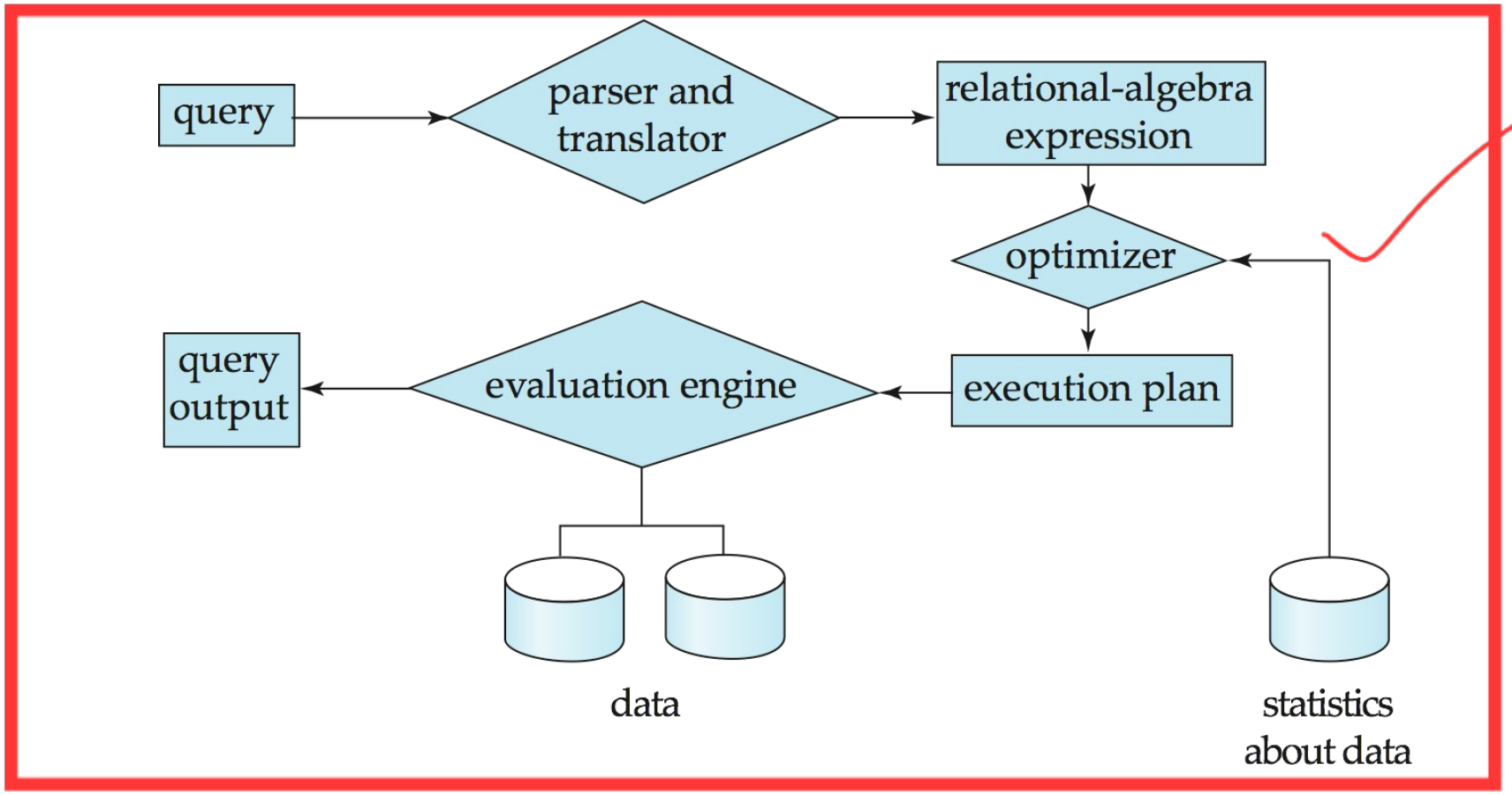


# Query Processing

8 )

> !

G 6



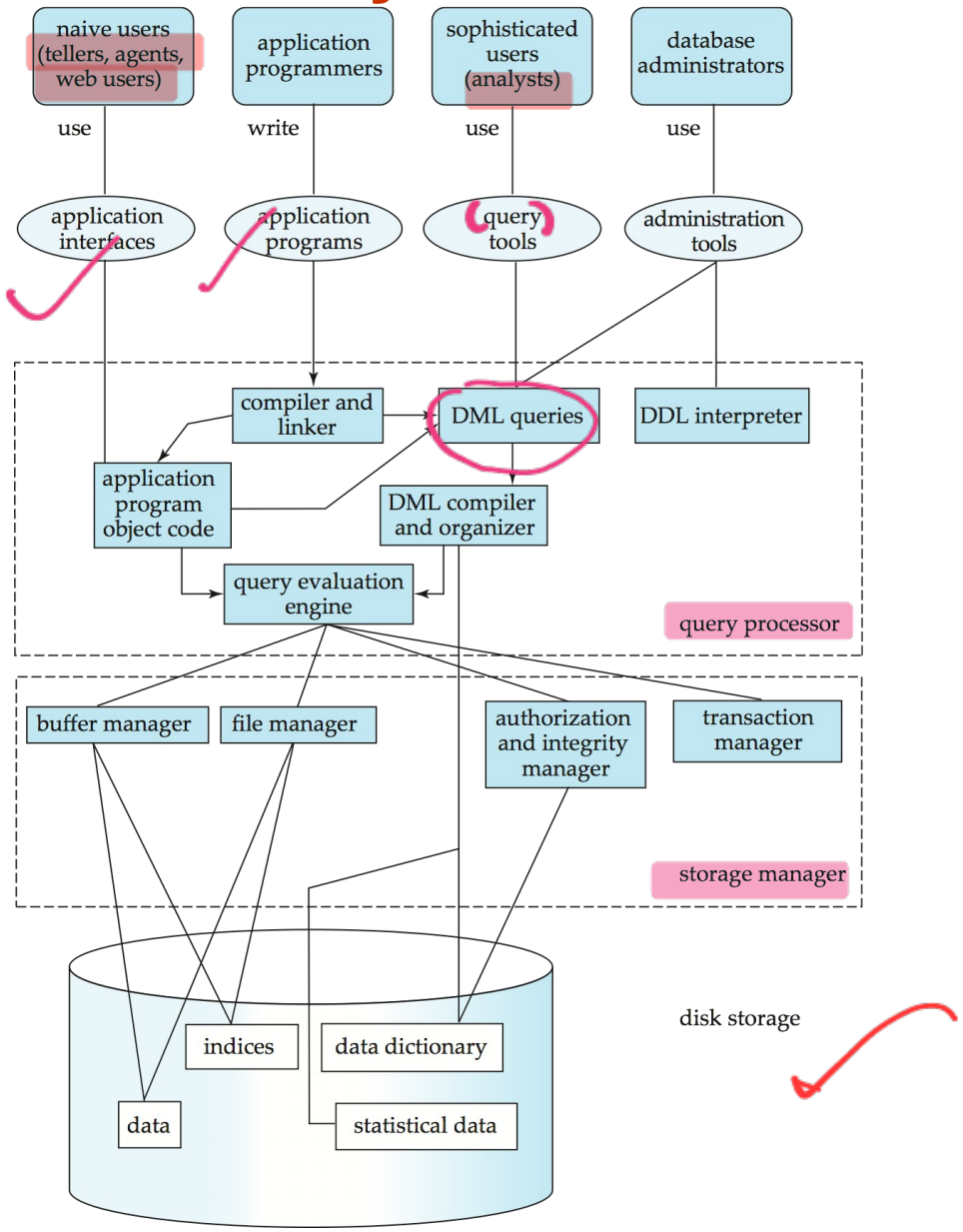


# Transaction Management

- K " l
- K " l
- transaction
- Transaction-management component
  % " *
- Concurrency-control manager
  "



# Database System Internals




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# Database Architecture

7 " " L "

" " %

n !

n :

n ) ' : *

n



# History of Database Systems

- 8M92 " 8MN2
  - 7 " @
  - )
- = 8MN2 8MJ2
  - # %
  - / % %
  - 7 ,
    - K % 7 % %
    - + & " & "
    - " + "
  - # : ' *



# History (cont.)

- 8M?2
  - &
  - )
  - ;
- 8MM2
  - =
  - =
  - 6
- 6 " >222
  - < =
  - 
- = >222
  - (
    (
    7
    O
    )/
    !



# End of Chapter 1



# Figure 1.02

| *ID* | *name* | *dept_name* | *salary* |
|---|---|---|---|
| 22222 | Einstein | Physics | 95000 |
| 12121 | Wu | Finance | 90000 |
| 32343 | El Said | History | 60000 |
| 45565 | Katz | Comp. Sci. | 75000 |
| 98345 | Kim | Elec. Eng. | 80000 |
| 76766 | Crick | Biology | 72000 |
| 10101 | Srinivasan | Comp. Sci. | 65000 |
| 58583 | Califieri | History | 62000 |
| 83821 | Brandt | Comp. Sci. | 92000 |
| 15151 | Mozart | Music | 40000 |
| 33456 | Gold | Physics | 87000 |
| 76543 | Singh | Finance | 80000 |

(a) The *instructor* table

| *dept_name* | *building* | *budget* |
|---|---|---|
| Comp. Sci. | Taylor | 100000 |
| Biology | Watson | 90000 |
| Elec. Eng. | Taylor | 85000 |
| Music | Packard | 80000 |
| Finance | Painter | 120000 |
| History | Painter | 50000 |
| Physics | Watson | 70000 |

(b) The *department* table


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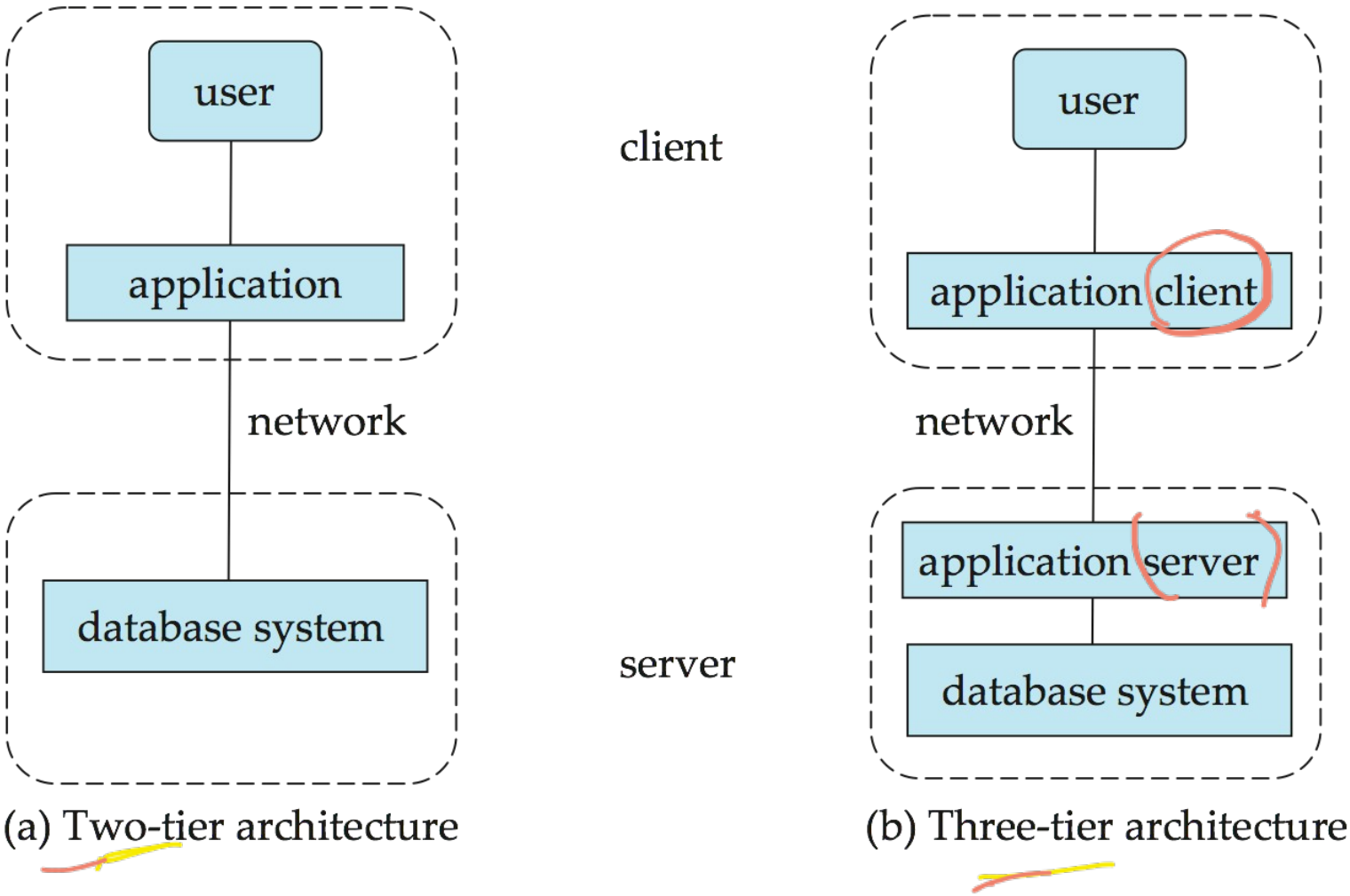


# Figure 1.04

| *ID* | *name* | *salary* | *dept_name* | *building* | *budget* |
|---|---|---|---|---|---|
| 22222 | Einstein | 95000 | Physics | Watson | 70000 |
| 12121 | Wu | 90000 | Finance | Painter | 120000 |
| 32343 | El Said | 60000 | History | Painter | 50000 |
| 45565 | Katz | 75000 | Comp. Sci. | Taylor | 100000 |
| 98345 | Kim | 80000 | Elec. Eng. | Taylor | 85000 |
| 76766 | Crick | 72000 | Biology | Watson | 90000 |
| 10101 | Srinivasan | 65000 | Comp. Sci. | Taylor | 100000 |
| 58583 | Califieri | 62000 | History | Painter | 50000 |
| 83821 | Brandt | 92000 | Comp. Sci | Taylor | 100000 |
| 15151 | Mozart | 40000 | Music | Packard | 80000 |
| 33456 | Gold | 87000 | Physics | Watson | 70000 |
| 76543 | Singh | 80000 | Finance | Painter | 120000 |


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# Figure 1.06




(a) Two-tier architecture

(b) Three-tier architecture